

Comments on Maryland RLA Regulation

Introduction

Thank you for developing proposed updates to the Code of Maryland Regulations (COMAR) to implement the requirements in law for Risk Limiting Audits (RLA).¹ We have experience researching, developing and supporting the implementation of RLAs in various states. Based on our experience we offer these comments.

From (affiliations listed for identification only):

Paul Burke, admin@VoteWell.net, Election Verification Network 2025 Research Award

Neal McBurnett, RLA Pioneer in Colorado

Stephanie Frank Singer, Campaign Scientific, former Chair, Philadelphia County (PA) Board of Elections

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¹mgaleg.maryland.gov/mgaweb/Laws/StatuteText?article=gel§ion=11-309&enactments=false

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- Draft rules are at https://dsd.maryland.gov/MDRIssues/5303/Assembled.aspx#_Toc221011335 and previous version at <https://mdsbe.app.box.com/s/j5vni3vds1qwd9uk3h2t4mk1vlfxiq5b/file/2075693997040>
- Sections not being amended are at <https://regs.maryland.gov/us/md/exec/comar/33.08.05>
- Law is at <https://mgaleg.maryland.gov/mgawebsite/Laws/StatuteText?article=gel§ion=11-309&enactments=false>
 - And law defines “voter–verifiable paper record” at <https://mgaleg.maryland.gov/mgawebsite/Laws/StatuteText?article=gel§ion=9-102&enactments=false>

Color Key for These Comments

Our suggested Additions are in red, underlined

Our suggested ~~Deletions are crossed out~~

[Brackets] show text which Maryland proposes to delete.

Comments are in blue italics

Big Issues

1. The regulation should provide criteria for how the State Board of Elections (SBE) selects contests, as the law requires

Maryland Election Law §9–102

(b)(1) Following each statewide election, the State Board, in collaboration with each local board, shall audit the accuracy of the voting system's tabulation of votes in each county by conducting a risk-limiting audit of:

- (i) at least one statewide contest; and
- (ii) any other contests selected for audit by the State Board...

(e)(1) The State Board shall adopt regulations to carry out this section.

- (2) The regulations adopted under paragraph (1) of this subsection to carry out risk-limiting audits in accordance with subsection (b) of this section **shall include:**
 - (i) **criteria for determining the contests to be audited;**

We recommend the regulations include:

- A. In 2026, one statewide contest, chosen either because of a very unexpected result or high public interest.
- B. In 2026, no other contest, unless there is a federal contest with a very unexpected result or high public interest.
- C. After 2026, the State Board will select all federal contests and one statewide contest as well as additional contests as feasible. When the voting system technology tracks the results of individual ballot sheets, it will be feasible to conduct ballot-level comparison audits and, thereby, audit many more contests. The State Board may select these additional contests randomly, based on unexpected results, public interest, or for any other reason.

Contest selection is critical because RLAs check the outcomes of the contests being audited (i.e. whether an accurate full hand count would have found the same winners in the contests being audited²). However, RLAs say nothing about the unaudited contests. For instance, if there were a ballot programming error where the credit for votes placed in a certain location on a ballot is given to the wrong candidate or if there were a hack of the voting system, only one contest, and not others, could be affected, as in Baltimore in 2020.³ Note that auditing additional contests in an RLA only adds incremental work to the batch tally or manual vote transcription process. The time-consuming set-up part of the audit, during which the specified ballots for the audit are located and pulled, is generally much more time consuming than recording votes for an additional contest.

The regulations also require an automated image audit, contracted to a third party, which so far has been Clear Ballot. The image audit checks all contests but relies on computer files of ballot images from the voting machines, so is not truly independent.⁴ Moreover, for Maryland's many BMD ballots, the current image audit uses barcodes printed on summary ballots. The barcodes cannot be verified by the voter.

Because of the different strengths and shortcomings of both types of audits, it is important to cover as many contests as feasible in both audits. It's important for the media, public and officials to know what the post election audits do and don't do, to prevent misinformation. A disclaimer in public reports and web pages will clarify what the audits do and don't show (suggested language is in 33.08.05.09 G 2 on pp.15-16),

2. The regulation should ensure that all the steps are conducted in the proper sequence:

- A. Publication of all software used for the RLA, including generating random numbers from a seed and RLA support software.
- B. Publication of the method for numbering batches and ballots in the manifest, so the randomly generated numbers correspond to one and only one batch or ballot.
- C. Ballot Reconciliation and Accounting audits.
- D. Checks of chain of custody, seal logs, etc.
- E. Publication of ballot manifest.
- F. Publication of preliminary results (such as batch subtotals and CVRs) corresponding to each item in the manifest, after any necessary consolidation to prevent linking votes to voters.
- G. Steps which can be done after the publication in step F and before the random draw in step H.
 - a. Independent accumulation of batch totals (or CVRs for ballot-level comparison audits) to check voting system totals.
 - b. Selection of contests.
- H. Initiation of the audit - - rolling of the dice for the generation of the random seed.
- I. Audit starts immediately after dice roll.
- J. Publication of the mapping from random selections to specific ballot sheets or batches in the manifest.
- K. Discrepancies between the CVR and ballot images should be accurately reported. If discrepancies can be attributed to errors introduced by the auditing process, these should be later enumerated. Discrepancies should not be "fixed" by the auditing staff without full disclosure.
- L. Post audit teams' batch tallies, and ballot sheet interpretations in ballot-level audits.
- M. Compare those posted results to preliminary voting system results and calculate if audit needs expansion.
- N. Expand audit if needed.
- O. Issue final audit report & findings, including lists of discrepancies.

² If paper ballots in storage were changed, added or removed, both an RLA and a hand count may come to the wrong conclusion.

³<https://www.baltimoresun.com/politics/elections/bs-md-pol-ballot-error-baltimore-district-1-20200603-n26t43fkmjadplqeqyb1oj4dki-story.html>

⁴ If the images are incomplete, overcomplete, or not a good representation of the paper, the audit may come to the wrong conclusion. There are examples of these problems "in the wild."

P. Certification.

3. Compliance audits are a key prerequisite for a meaningful RLA or automated audit.

"If the compliance audit does not generate convincing affirmative evidence that the ballots have not been altered and that no ballots have been added or lost, a risk-limiting audit may be mere theater".⁵

Maryland conducts some important compliance audits such as a reconciliation audit, which compares the number of voters checked-in to the number of ballots cast in a polling place. Other compliance audits are needed to provide evidence that the ballots have been safeguarded throughout the process and thus provide reliable evidence. We recommend exploration of additional steps in the future such as digital signatures on each batch of scanned ballots, and ways to increase the independence of the auditors. (See 33.08.05.09 H on p.16.)

The paper ballots are the evidence for RLAs, so if the ballots are not properly preserved, the RLA is not reliable. The scanned images are the evidence for the automated software audit, so if the images are not properly created and delivered, the audit is not reliable.

4. Certain technical issues with risk-limiting audits should be clarified in the regulation:

- A. The random seed should be generated using dice. This method generates a truly random number in a publicly observable way. The RLA needs to start immediately after the dice roll, so no skeptic thinks the sample could be "fixed".⁶
- B. The proposed rule has batch comparison RLAs as the default. We suggest adding permissive wording to let the State Administrator authorize other methods. Other methods might be useful in the future.
- C. The call-and-tally method (using four-person teams: a caller, a verifier, and two talliers) should always be used, not the sort and stack method. It requires learning only one method, since it is faster for multiple contests and no slower for one contest. Research has found call-and-tally to be more accurate.⁷
- D. The regulation should be clear that the RLA stops only when either the risk limit is met or when a full manual recount has been completed. The RLA may not change the outcome unless a full manual recount has been completed. The law says:

manually examine randomly chosen individual voter-verifiable paper records or batches of voter-verifiable paper records until the risk limit is met or the correct election outcome is established;

Implementing better post-election audits, including RLAs, is a big step forward for Maryland. We congratulate Maryland on its past work, and hope you will find these comments useful.

⁵ <https://www.stat.berkeley.edu/~stark/Preprints/gentle12.pdf> See also <https://georgetownlawtechreview.org/wp-content/uploads/2020/07/4.2-p523-541-Appel-Stark.pdf>

⁶ Cleveland, OH, staff pled no contest to fixing the ballot samples before an audit. https://www.cleveland.com/metro/2007/11/elections_board_workers_to_tak.html For a recount, the Registrar of Voters in Las Vegas, NV, opened ballot boxes in secret, did a secret recount before the public one, and hid the results. https://votewell.homestead.com/clark_county_report.pdf

⁷ 0.48% error in read and mark; 2.13% error in sort and stack, on p.46 of Goggin, Byrne and Gilbert, (March 2012). ["Post-Election Auditing: Effects of Procedure and Ballot Type on Manual Counting Accuracy, Efficiency, and Auditor Satisfaction and Confidence"](#)

COMAR Changes Proposed by Board 12/18/25, published 2/6/26 with our suggestions added

Title 33 STATE BOARD OF ELECTIONS

Subtitle 07 ELECTION DAY ACTIVITIES Chapter 10 Non-Voting Hours Procedures

Authority: Election Law Article, §§2-102(b)(4)[and], 2-202(b) and 11-309, Annotated Code of Maryland

.01 Ballot Box Return and Inspection.

A. Numbered Seals Required. Any seal used by a local board shall be numbered.

B. Reception Team to Receive. The sealed [or locked] ballot boxes from the polling places shall be received by a reception team.

C. Marking.

(1) Each ballot box shall be marked as having been inspected before it is delivered to a ballot reception team; and

(2) Each ballot box shall be marked with the batch label prior to certification in D.

D. Inspection and Certification.

(1) The reception team shall:

(a) Inspect each ballot box and its seal [or lock]; and

(b) Prepare and sign a certification of the box's condition.

(2) The certification for each box shall include:

(a) The precinct number;

(b) The seal number[, if a seal is used];

(c) The condition of seal [or lock]; [and]

(d) Any defects observed[.]; and

(e) The batch label.

(f) The names and signatures of the two-person bipartisan team who picked up and transported the ballot box to the local board of elections. *This is important for chain of custody.*

Canvassing Chapter 05 Post Election Verification and Audit

Authority: Election Law Article, §§11-309, Annotated Code of Maryland

.01 Definitions.

A. In this chapter, the following terms have the meanings indicated.

B. Terms Defined.

(1) "Automated software audit" [is a software audit performed by an entity other than the vendor of the certified voting system] has the meaning stated in Election Law Article, §11-309, Annotated Code of Maryland. *The law states: " 'Automated software audit' means an audit of electronic images of ballots cast in an election using software that is independent of the voting system." Note that an automated software audit which uses the voting machine vendor's algorithm for interpreting barcodes should NOT be considered as "using software that is independent of the voting system."⁸ Using the voting system's images is also a dependency, so images must be checked.*

⁸ In past elections, Maryland's automated audit vendor, Clear Ballot has used the voting system's algorithm for interpreting barcodes. Not using barcodes means interpreting text on BMD ballots by OCR, as Clear Ballot offered in Georgia, and Enhanced Voting and AuditEngine have done in Georgia.

(2) "Ballot-level comparison audit" means a type of risk-limiting audit in which the local board examines voter selections on randomly selected individual paper ballots and compares them with corresponding cast vote records.

(3) "Ballot polling audit" means a type of risk-limiting audit in which the local board examines voter selections on randomly selected ballots and compares the results from all the sampled ballots to the published results.

(4) "Ballot manifest" means a detailed catalog prepared by the local board listing the batch label, number of ballots, storage location, and any other material required by the State Board as necessary to identify ballots for the risk-limiting audit.

(5) "Batch" means a grouping of ballots by criteria determined by the State Board and individually stored together in one labeled container ~~in a ballot box.~~

(6) "Batch Comparison audit" means a type of risk-limiting audit in which the local board examines and tabulates selections on randomly selected batches of ballots and compares the sums to the published counts.

(7) "Batch Label" means the uniform naming convention determined by the State Board that is used to identify the batch of ballots.

(8) "Cast Vote Records" means an auditable document or electronic record that purports to reflect the selections ~~a voter~~ made on a ballot.

[(2)] (9) "Discrepancy" means the difference between the preliminary published voting system results and the results of an automated or [manual] risk-limiting audit.

[(3)] (10) "Manual count [audit]" has the meaning stated in Election Law Article, §11-309, Annotated Code of Maryland.⁹

[(4)] ~~(11) "Precinct" includes an early voting center in Regulations .02 — .06 of this chapter.~~

[(4)] (11) "Polling place" includes election day voting locations and early voting centers.

The current regulation's conflation of early voting center and precinct is not needed¹⁰ and is confusing. The definition of "voting center" and "precinct" is different. Early voting centers service voters from many different precincts. Making new definitions for commonly used words in a regulation is confusing and bad practice.

[(5)] "Previous comparable general election" has the meaning stated in Election Law Article, §11-309, Annotated Code of Maryland.]

⁹ The law says "Manual count" means inspection of voter-verifiable paper records by hand and eye to obtain vote totals in a contest.

¹⁰ Conflating "princinct" and "Early Voting Centers" for regulations .02 -.06 is not needed:

- .02 deals with the purpose of audits so this conflation is not needed.
- .03 does not use the word "precinct" so this conflation is not needed.
- .04 & .05 have been about selecting precincts for optical scanner tapes on election day at each polling place, compared against the voting system numbers and checking the summation. Early voting sites close before election day, so should not create scanner tapes. If the EMS generates totals tapes later, there is little use in comparing them to the EMS. In any case, the checking of poll tapes against the EMS should be a 100% verification, not based on samples.

(12) "Random seed" means a randomly generated number that is used to generate a random number sequence to select ballot sheets or batches of ballots for audit.

(13) "Risk Limit" has the meaning stated in Election Law Article, §11-309, Annotated Code of Maryland.¹¹

(14) "Risk-Limiting Audit" has the meaning stated in Election Law Article, §11-309, Annotated Code of Maryland.¹²

(15) "Risk-limiting audit software" means the software used by the State Board in order to conduct the risk-limiting audit.

[(6)] (16) "Voter-verifiable paper record" has the meaning stated in Election Law Article, §9-102, Annotated Code of Maryland.

.06 Post-Election Audit-~~Ballot Reconciliation and Ballot Accounting~~ Precincts-

A "precinct audit" is not a common term to describe this type of audit. The common term for what is described is a "Ballot Reconciliation and Ballot Accounting Audit". Prior to a post-election-tabulation-audit there should be two types of audits:

- *A ballot reconciliation audit where the number of voters who checked in is compared to the number of votes cast.*
- *A ballot accounting audit where the number of printed blank ballots is compared to the number of voted ballots and the number of remaining ballots at each polling place. This makes sure that no ballots were added or subtracted.*

See:

- https://www.eac.gov/sites/default/files/electionofficials/QuickStartGuides/Ballot_Reconciliation_Election%20Day_EAC_Quick_Start_Guide_508.pdf
- <https://electionsgroup.com/resource/ballot-reconciliation/>
- <https://sosmt.gov/wp-content/uploads/Ballot-Reconciliation-Process-and-Checklist.pdf>
- <https://www.brennancenter.org/sites/default/files/legacy/Democracy/9.25.08.BestPracticesChecklist.pdf>

A. [Automated] ~~Automatic Audit Procedures.~~ Ballot Reconciliation Procedures.

The words Automatic Audit are too similar to the "Automated Software Audit" which is not addressed here. Ballot reconciliation is key for both the RLA and the Automated software audit. Prior to any tabulation audit (Automatic or RLA)

(1) ~~(text unchanged)~~ For every polling place ~~precinct~~ in an election, the State Administrator shall produce a ballot reconciliation report with the following information.

- (a) The number of voters who were checked-in to vote as reported by the electronic pollbooks;
- (b) The number of ballots (or ballot first sheets for multisheet ballots) cast as reported in ~~the Election Management System~~ the voting system.
- (d) The difference, if any, between the number of voters checked-in to vote and ballots cast.

¹¹ The law states: "Risk limit" means the small, predetermined maximum chance that a risk-limiting audit will not require a full manual count of voter-verifiable paper records in an audited contest if a full manual count of the voter-verifiable paper records would find a different outcome than the outcome determined by the electronic count.

¹² The law states: "Risk-limiting audit" means a postelection audit procedure that employs statistical methods to ensure a large, predetermined minimum chance of requiring a full manual count of voter-verifiable paper records in an audited contest if a full manual count of the voter-verifiable paper records would find a different outcome than the outcome determined by the electronic count.

(2) The election director shall conduct the audit procedures under §B of this regulation for each batches of ballots scanned at that polling place where precinct if the State Administrator's ballot reconciliation report indicates that the difference between the number of voters checked-in to vote and the number of ballots cast is five or more.

~~B. Automatic Audit Procedures—Addition of Batches to Risk Limiting Audit. (1) Batches shall be added to risk-limiting audit under COMAR 33.08.05.09. The election director shall~~

~~(a) add the batches identified in §A of this regulation to the batches audited by the risk-limiting audit; and •~~

~~(b) conduct the audit procedures under this section for each batch of ballots cast at precinct identified under §A of this regulation. *Paragraph B cannot be changed in this way. The old paragraph B had instructions to deal with extra and missing ballots, which we update below. Furthermore, adding batches to the RLA does not resolve mismatches.*~~

~~{~~B. Manual. **Adjustments for ballot count discrepancies**

(1) The election director shall conduct the audit procedures under this section for each polling place ~~precinct~~:

(a) With batches or ballots that were selected for the RLA under Regulation .04 of this chapter; and

(b) Identified under §A of this regulation.

(2) For each polling place ~~precinct being audited~~ identified in B(1), the election director shall:

(a) Hand count the signed voter authority cards associated with each voting unit ~~in the precinct~~; and

(b) Compare the number of signed voter authority cards against the number of ballots cast at each voting machine as reported by the voting system. ~~as reported in the Election Management System.~~

(c) Compare the number of voters who voted at all of the voting machines to the number of voters who checked-in according to the pollbooks.

(3) If the number of voters who checked in does not match the number of ballots cast, the election director shall:

(a) Review other election materials from the polling place ~~precinct~~ to determine the cause of the difference; and

(b) Report to the State Administrator the explanation for the difference.

(4) If the polling place has fewer ballots than pollbook check-ins, the election director will determine which ballot styles are missing. The RLA escalation rules must assume each missing ballot had votes for the runners-up in every contest being audited.¹³

(5) If the polling place has more ballots than pollbook check-ins in any style, the election director will determine which ballot styles have extra ballots. The RLA escalation rules must assume any votes for winners on styles with extra ballots are erroneous, up to the number of extra ballots in that style.¹⁴

(6) Determine if a full hand count would address any remaining problems†

C. (text unchanged) Notification.

The election director shall report to the local board and the State Administrator the results of the audit.

¹³ Dr. Stark calls these zombie ballots. <https://www.stat.berkeley.edu/~stark/Preprints/2ez12.pdf> See https://dl.acm.org/doi/10.1007/978-3-030-54455-3_23 for a more thorough treatment.

¹⁴ May require SHANGRLA. <https://arxiv.org/abs/2207.01362> Examples at https://docs.google.com/spreadsheets/d/1YUaPR_g-tTyaTtaXq5w26-UfvaumXvAFcbrnzIXOmCA/edit?gid=0#gid=0

{D. Certifying the Election.

If the reconciliation audit demonstrates an unexplained discrepancy between the number of checked-in voters~~signed voter authority cards~~ and the number of ballots cast and that discrepancy is: (1) Less than the vote margin of any contest on the ballot in that precinct, the local board of canvassers shall accept the audit results and allow for the certification of ~~certify~~ the election; or (2) Greater than the vote margin of any contest on the ballot in that precinct, the local board of canvassers may not certify the election until any corrective actions required by the State Administrator are completed.]
Paragraph D cannot be deleted since mismatch needs to be addressed for non-RLA contests.

E. For each polling place, the election director shall send the State Administrator a ballot accounting report which compares the number of printed ballots delivered to each polling place to the total of the number of ballots voted, the number of ballots voided and the number of printed ballots remaining at the end of voting. The election director will investigate discrepancies of 5 or more and report findings and recommendations to the State Administrator.

.08 Post-Election Audit — Ballot Tabulation Audit — In General.

- A. Audits Conducted by the State Administrator. The State Administrator shall conduct an automated software audit of the electronic images of all ballots cast:
- (1) After each Statewide primary election; and
 - (2) After each Statewide general election.
- B. Audit Conducted by the Local Boards. Each local board shall conduct a [manual] risk limiting audit of voter-verifiable paper records[:] after each Statewide election.
- [(1) After each Statewide general election; and
 - (2) After a Statewide primary election at the direction of the State Administrator.]

C. Reporting of Audit Results.

- (1) [Within 14 days after the conclusion of the manual audit, the] Prior to certification of the election by the Board of State Canvassers. ~~the~~ State Administrator shall post on the State Board website:
 - [(a) The precincts and number of votes selected for the manual audit in each county and the manner in which the precincts and votes were selected; (b) The results of the manual audit; and (c) Any discrepancy shown by the manual audit and how the discrepancy was resolved.]
 - (a) ~~Prior to certification of the election by the Board of State Canvassers,~~ Ihe results of the risk-limiting audit, including:
 - (i) an overview of the risk-limiting audit process; and
 - (ii) an explanation ~~or~~ and video of the batch or ballot selection process.
 - (b) ~~Within 1 week after certification of election results by the Board of State Canvassers,~~ the results of the automated software audit, including:
 - (i) An overview of the automated audit process;
 - (ii) The audit reports generated for each county; and
 - (iii) An explanation of any discrepancy greater than 0.5 percent of total votes cast in any given contest; and
 - (iv) Any additional steps taken to resolve any discrepancy.
- [(2) Before the Board of State Canvassers certifies the results of an election, the State Administrator shall post on the website information about the automated audit, including: (a) An overview of the automated audit process; (b) The audit reports generated for each county; and (c) An explanation of any discrepancy greater than 0.5 percent of total votes cast in any given contest; and (d) Any additional steps taken to resolve any discrepancy.]
- (2) Prior to certification of the election by the Board of State Canvassers, the State Administrator shall report the results of the risk-limiting audit and the automated software audit to the Board of State Canvassers and the State Board.

.09 Post-Election Audit — Ballot Tabulation Audit — Risk-Limiting Audit.

A. In General

(1) Audit Method.

(a) Except as provided in (b), a batch comparison audit shall be conducted in accordance with Election Law Article, §11-309, Annotated Code of Maryland.

(b) Exception to batch comparison. Given a jurisdiction's voting machine technology, or given time constraints, if the State Administrator may determine that, ~~a batch comparison audit cannot be completed prior to the Board of State Canvassers certification~~ the audit may be conducted using the following methods:

(i) Ballot-level comparison;

(ii) ballot polling; or

(iii) a hybrid of batch comparison, ballot-level comparison, or ballot polling.

(iv) the ONEAudit method¹⁵

Note: ballot-level comparison is more efficient than batch comparison but is most practical when the voting machines have the ability to affix a unique number on the ballot (but after the ballot is cast so that it cannot be associated with a voter). Ballot polling tends to require very large samples when the margin of victory is even moderately small. For instance for a margin of 1% in a two-candidate plurality contest, the expected sample size for a 10% risk limit is on the order of 30,000 ballot sheets. Moreover when discrepancies are found, the cause is difficult to determine.

(2) Risk limit. The risk limit shall be 10 percent.

(2.5) Conflicts of interest

(a) Audit staff must disclose any personal or financial ties to candidates, campaigns, or vendors involved in the election, thereby helping to ensure that audits are conducted impartially and ethically.¹⁶

(3) Timing of Audit

- a. The State Administrator publishes all software used for the RLA, including the generation of random numbers from a seed and RLA support software.
- b. The State Administrator publishes the method for numbering batches and ballots in the manifest, so the randomly generated numbers correspond to one and only one batch or ballot.
- c. The Ballot Reconciliation and Accounting audits are conducted.
- d. The election director checks chain of custody and seal logs.
- e. ~~(a)~~ No later than 17 days after election day, the local board shall provide to the State Board a ballot manifest, in the manner prescribed by the State Board, which shall publish it.
- f. Before the sample is chosen, the local board shall load into the risk-limiting audit software the total vote counts from the EMS and the vote counts for each batch eligible to be sampled for the batch audit, and the CVRs for each CVR eligible to be sampled for the ballot-level comparison audit. These individual vote counts shall be accumulated, independently of the EMS, to check the EMS totals.

¹⁵ <https://arxiv.org/abs/2303.03335> and <https://arxiv.org/abs/2507.22179>

¹⁶ Quote from Ethics standard of https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fwww.eac.gov%2Fsites%2Fdefault%2Ffiles%2F2026-02%2FDraft_Audit_Standards_021826_Clean.docx&wdOrigin=BROWSELINK

- g. Before the sample is chosen, the State board should publish the preliminary results on the website, after any necessary consolidation to prevent linking votes to voters.
- h. The state board shall select contests to be audited.
- i. Initiation of the audit. The State Administrator shall manage rolling of the dice for the generation of the random seed.
- j. Retrieval of ballots and tallying start immediately after dice roll.
- k. The State Administrator publishes the mapping from random selections to specific ballot sheets or batches in the manifest.
- l. Post audit teams' batch tallies, and ballot sheet interpretations in ballot-level audits
- m. Compare those posted results to preliminary voting system results, note discrepancies, and calculate if audit needs expansion.
- n. Expand audit if needed.
- o. The State Administrator issues a final audit report & findings, including lists of discrepancies.
- p. ~~(b)~~ The local board shall commence a local risk-limiting audit no sooner than 21 days after election day; and *[Are we right this 21-day rule only applies to local RLAs?]*
- q. ~~(c)~~ Complete a local risk-limiting audit no later than two days before the Board of State Canvassers convenes under Election Law Article, §11-503(a)(1), Annotated Code of Maryland
- r. Local and State Boards certify the election.

B. Public Notice and Observation.

- (1) A local board shall:
 - (a) At least 3 days before the risk-limiting audit starts, provide notice of the risk-limiting audit by:
 - (i) Sending the notice to the chairman of the county central committee, each candidate for the contest to be audited who is not a candidate of a principal political party, and the State Administrator;
 - (ii) Posting the notice on its website; and
 - (iii) Posting the notice in a prominent and publicly accessible location at its office.
 - (b) Allow, to the extent practicable, for public observation of each part of the manual count of ballots of the risk-limiting audit process, close enough to read the ballot marks and tally sheets, or with unrecorded projection of them.
- (2) The State Board shall:
 - (a) Provide at least 3 days notice of the time, date, and location for conducting risk-limiting audit initiation procedures required in §E of this Regulation by posting the notice on its website.
 - (b) Allow, to the extent practicable, for in-person and remote public observation of risk-limiting audit initiation procedures.

C. Selection of Contests for Audit.

- (1) Determination of ~~number of~~ Statewide and Federal contests. After publication of preliminary totals, and no later than the day before the start of the risk-limiting audit, the State Board shall determine the ~~number of~~ contests to be audited as follows:
 - ~~(a) Consider the number of statewide contests in the election;~~

(b) Consider local boards' resources, including provisional processing, canvass, and recounts;

(c) In primary and general elections, select all statewide and federal contests with very unexpected preliminary results.

(d) In 2026, if no statewide contest was selected under (c), then select the statewide contest with the most public interest or the top of the ballot.

(e) After 2026, select all federal contests, the statewide contest with the most public interest or the highest on the ballot, as well as additional contests as feasible. The State Board may select these additional contests randomly, based on public interest, or for any other reason.¹⁷

~~(e)~~ (f) For a primary election:

(i) If one contest is selected, require auditing of at least one statewide contest on one party's ballot.

(ii) If two contests are selected, require auditing of at least one statewide contest on each party's ballot.

(g) When the winning margin is so small that an RLA sample is not practical, the State Administrator may order a 100% hand recount instead of the RLA sample.

~~(d) In a general election, require auditing of at least one statewide contest~~

~~(2) Determination of Statewide Contest. No later than the day before the start of the risk-limiting audit, the State Board shall randomly select from all statewide contests at least one contest to be audited.~~

(3) Requests by Local Boards to Audit Local Contests.

(a) A local board may conduct a risk-limiting audit of one or more local contests at the discretion of the local board.

(b) If a local board conducts a risk-limiting audit of a local contest:

(i) All statewide contests must be completed prior to the commencement of the audit; and

(ii) The timing of completion of the risk-limiting audit of the statewide contest may not be affected by the local contest audit.

(c) A local board may request the State Administrator's approval, in writing ~~no later than 1 month prior to the election~~, to use the risk-limiting audit software to conduct the local contest audit.

If a local contest has a very unexpected result and the local board wants to audit it, the county should be allowed to use the software, if allowed by contract, even if the county had not made a previous request.

D. Local Board Preparations for Audit

(1) Ballot control. Each local board shall store ballots as follows:

(a) In a sealed ballot box, with only ballots from the same batch; and

(b) With the batch label on the exterior of the ballot box, which identifies the batch as it appears on the ballot manifest.

(2) Requirement to provide ballot manifest. Each local board shall provide to the State Board no later than 17 days after election day:

(a) a ballot manifest in the manner prescribed by the State Board; *(Already stated in A(3)(a))*

(b) an estimate of all provisional and mail-in ballots that have not yet been canvassed; and

(c) certification that all batches listed on the ballot manifest have been verified by the local board as required by the State Board.

¹⁷ *When Maryland's voting system technology facilitates ballot-level comparison audits, it will be feasible to audit many more contests.*

E. Initiation of Risk-Limiting Audit.

On the ~~day before~~ morning of the first day of the risk limiting audit, the State Administrator shall hold a public, videoed meeting during which the State Administrator shall:

- a) Select random choice of a seed number to input into risk-limiting audit software by rolling dice; and
- b) ~~(2)~~ Using the randomly-selected seed, risk limit, ~~and~~ margin of victory for the contest or contests selected to be audited, and the ballot manifest, determine the batches or individual ballots to be audited by each local board.

F. Manual Count and Escalation Procedures.¹⁸

(1) Each election director shall:

- (a) Prepare for a risk-limiting audit by recruiting and training manual count teams;
- ~~(b) Instruct teams to use the sort method described in subsection (2) of this Regulation if one Statewide contest are audited;~~
- (b) Instruct teams to use the tally method described in subsection ~~(3)~~ (2) of this Regulation. ~~if more than one Statewide or local contest is audited;~~ ~~(d)~~
- (c) Issue the teams batches of ballots or voter-verifiable paper records individual ballots as specified by the State Administrator per E(2)(b) chosen by the risklimiting software for manual count; and
- (d) Ensure that the count teams have no knowledge of the preliminary published voting system results for the batches or individual ballots being audited.
- (e) Record in the audit log:
 - (i) The team identifier;
 - (ii) The ballots ~~voter-verifiable paper records~~ issued to the team; and
 - (iii) Upon completion, ~~the voter-verifiable paper records~~ ballots returned by the team.
- (f) For batch-level comparison audits, enter the vote totals reported by manual count teams on the consolidated tally sheets and complete and sign the tally sheet and enter the results into the risk-limiting software.
- (g) Have software or staff other than the counting team members compare the results of the manual count against the preliminary published voting system results for those voter-verifiable paper records;
- (h) If there are any ~~inaccuracies~~, discrepancies,

¹⁸ The RLA law states:

(4) A risk-limiting audit shall:

(i) manually examine randomly chosen individual voter-verifiable paper records or batches of voter-verifiable paper records until the risk limit is met or the correct election outcome is established;

(ii) be completed before the Board of State Canvassers transmits a certified statement of the election results to the State Board under § 11-503(a) of this title; and

(iii) be observable by the public to the maximum extent practicable.

(5)(i) If a risk-limiting audit finds that the election outcome determined by the electronic count is inaccurate, the State Administrator, in consultation with the State Board, shall direct the relevant board of canvassers to immediately investigate the matter to determine an accurate election result.

(ii) The board of canvassers shall conclude the investigation within 3 days.

(iii) The board of canvassers shall correct the election result in accordance with regulations adopted by the State Board.

(6) The State Administrator shall report the results of the risk-limiting audit to the Board of State Canvassers and the State Board.

<https://mgaleg.maryland.gov/mgaweb/Laws/StatuteText?article=gel§ion=11-309&enactments=False&archived=False>

- (i) ~~retally~~ retabulate the voter verifiable paper records; using a team without knowledge of the voting machine results
- (ii) When widespread errors are detected, such as configuration errors that may affect many ballots, consider doing a 100% hand recount, or rescanning with correct configurations and restarting the audit.
- ~~(i) (iii) If manual and voting system counts agree, and enter~~ Complete and sign the re-tally sheet and enter the results into the risk-limiting software.
- ~~(j) Within 24 hours after completion of the audit, submit the results to the State Administrator.~~
- (i) If the risk-limiting software finds that the chance that a full manual count would find a different outcome is less than the risk limit, then the audit is over, and the State Administrator must be notified within 24 hours.
- (j) If the risk-limiting software finds that the chance that a full manual count would find a different outcome is more than the risk limit, then the audit is not over and more batches and ballots must be audited until either:
 - (i) the risk-limiting software finds that the chance that a full manual count would find a different outcome is less than the risk limit. In this case, the audit is over, and the results must be submitted to the State Administrator within 24 hours.
 - Or
 - (ii) a full manual recount has been conducted.

~~Within 24 hours after completion of the audit, submit the results to the State Administrator.~~

- ~~(2) Sort Method. Each manual count team shall: (a) Designate one team member who shall sort and a second team member who shall watch to ensure accuracy; (b) Sort all assigned voter verifiable paper records into one of the following batches: (i) A batch for each candidate or ballot question response for a contest being audited; (ii) A batch for voter verifiable paper records without a vote for a contest being audited; (iii) A batch for each officially filed write-in candidate for a contest being audited; and (iv) A batch for all other write-in votes; (c) If team members do not agree on how a vote should be counted: (i) Refer the voter verifiable paper record to the election director; and (ii) For auditing purposes only, accept the election director's determination of how the vote shall be counted; (d) Once all of the voter verifiable paper records have been sorted, ensure that each team member independently counts the voter verifiable paper records in each batch; (e) If the team members' results are not identical, retabulate the voter verifiable paper records until obtaining identical results; and (f) When the team members' results are identical: (i) Record the vote totals on the batch tally sheet; (ii) Sign the batch tally sheet; and (iii) Give the batch tally sheet and the voter verifiable paper records to the election director.~~
- (32) Tally Method, which can be used for batch and ballot-level comparison audits. Each manual count team shall:
 - (a) Include one caller, one watcher, and two tally clerks.
 - (b) For each voter verifiable paper record assigned to it:
 - (i) Have its caller call the votes cast in the contests being audited; (ii) Have its watcher ensure the accuracy of the calling; and
 - (iii) Have its two tally clerks independently record the votes as they are called. For ballot-level comparisons, one tally clerk enters on paper, and the other into the risk-limiting audit software.
 - (c) Ensure that its tally clerks periodically compare their results to each other's;
 - (d) If the tally clerks' results are not identical, retabulate voter verifiable paper records beginning with the point of the last successful comparison check, and repeat until the two tally clerks obtain identical results; and
 - (e) When all votes assigned to the team have been tallied, ensure that its tally clerks:

- (i) Record the vote totals on the ~~batch~~ tally sheet;
- (ii) Sign the ~~batch~~ tally sheet; and
- (iii) Give the ~~batch~~ tally sheet and the voter verifiable paper records to the election director for entry into software if not already done

(43) Full manual recount~~Resolving Electronic Count.~~

- (a) If the full manual count of results in an audited contest confirms the preliminary published outcome, then the audit is over.
- (b) If the full manual count of results in an audited contest does not confirm the preliminary published outcome ~~electronic count of results~~ for that contest, the local board of canvassers shall investigate the matter to determine an accurate result, including:
 - (i) A review of all tally sheets and other documentary materials from the risk-limiting audit;
 - (ii) A review of all canvass minutes;
 - (iii) Interviews with the election director; and
 - (iv) Any other information that would explain the difference between the audit and the ~~electronic count~~ preliminary published voting system results.
- (c) No later than 3 days after beginning the investigation, the local board of canvassers shall:
 - (i) Provide to the State Administrator written findings explaining the difference; and
 - (ii) If it concludes that the preliminary published voting system results ~~electronic count~~ is inaccurate, adopt corrections to certified results in accordance with the full manual recount and Election Law §§ 11-308 and 11-401, Annotated Code Maryland

G. Public Information

(1) Issue final audit report & findings, including lists of discrepancies.

(2) Reports and websites referring to audit results will note these limitations:

- a) The risk-limiting audit (RLA) has findings about the contests it audits. It says nothing about the unaudited contests. It's reliable to the extent all paper ballots are tracked and protected from changes.
- b) The automated software audit is reliable to the extent the original scans from the voting system were accurate, except that it depends on barcodes printed by the voting system, which the voter cannot verify. [Include the "except..." clause until and unless the software audit uses OCR instead of bar codes]
- c) The scope of the RLA and automated software audit is to check the outcome of the tabulation. Other issues, such as eligibility of the voters and tracking of ballots are checked by careful election handling, usually by bipartisan teams.

H. Exploring Steps to Improve Audits in Future

The Administrator and Election Directors are authorized to take the following steps and report results to the State Board after each election cycle.

1. Steps can be taken to ensure independence in audits. An EAC draft says:¹⁹

- a. In many cases, supervisors audit the work of their subordinates, which creates a degree of independence between the auditor and those who performed the original tasks.

¹⁹https://view.officeapps.live.com/op/view.aspx?src=https%3A%2F%2Fwww.eac.gov%2Fsites%2Fdefault%2Ffiles%2F2026-02%2FDraft_Audit_Standards_021826_Clean.docx&wdOrigin=BROWSELINK

Similarly, state election offices often audit the work of local jurisdictions, maintaining a clear distinction between the audited entity and the auditor. Local jurisdictions might promote independence by involving officials from different departments to oversee election audits.

- b. Additionally, statutory oversight bodies or external auditors, who are credentialed to assess government entities, may be engaged to conduct audits. These types of measures can help ensure that audit teams are appropriately distanced from the specific election processes or outcomes under review, thereby upholding the standard of independence and fostering public trust in audit results.

For instance, the tallying portion of the audit can be conducted by staff from another county.

2. “Generate convincing affirmative evidence that the ballots have not been altered.”²⁰ and that their images are accurate, by a series of quality control steps:
 - a. Obtain the hash or digital signature²¹ of each scanned ballot image file and post them on a time-stamped independent archive not under the control of officials, as soon as each batch is scanned.
 - b. Randomly select 1 or more ballots from each batch.²² Publicly compare them to their scanned images, as soon as each batch is scanned, before the ballots are stored out of sight. If the images don’t match the ballots, the scanner needs to be readjusted or replaced, the batch needs to be re-scanned, the new hash/signature posted, and new random ballots compared.
 - c. Compare ballots in the risk-limiting audit sample to the verified, hashed or signed image file, and report and investigate any discrepancies in order to provide evidence that the ballots have not been altered.
3. Track weights²³ or counts of ballots and envelopes at each stage of processing, from mail or printer to scanning, to be posted on a time-stamped independent archive, with reconciliation of what happened to all ballots in order to provide evidence that no ballots have been added or lost.
4. Have an independent auditor send thank you cards to a random sample of voters for whom an absentee ballot is received, with instructions how to reply if they didn’t send the ballot. Any such replies should be investigated and reported.²⁴
5. Add a message in ballot tracking messages, if not already done, saying how to reply if the voter didn’t send the ballot.

²⁰ Stark & Lindeman wrote, “If the compliance audit does not generate convincing affirmative evidence that the ballots have not been altered and that no ballots have been added or lost, a risk-limiting audit may be mere theater” <https://www.stat.berkeley.edu/~stark/Preprints/gentle12.pdf>

²¹ Hashes & digital signatures are explained at <https://docs.google.com/document/d/1teEGQjwUmv1v5APG6zqrPsO580ozBG6AlbQezcmxQW8/edit?tab=t.0#heading=h.gr9zoc5laewz>

²² Exploration can start small. As in an RLA, small samples cost less but risk missing more issues. The goal is to have enough sample overall in the jurisdiction as a whole, not a representative sample of each batch. The same sample size calculation used below for thank you cards can be used or adapted.

²³ SC and RI have used weights to count ballots. <https://scvotes.gov/wp-content/uploads/2024/12/FinalRpt9.pdf> and <https://verifiedvoting.org/wp-content/uploads/2020/07/RI-RLA-Report-2020.pdf>

²⁴ The statewide sample should be enough so the thank you card sample has 95% chance of detecting a statewide forged ballot rate of 1% (statewide 300 cards= $\log(0.05)/\log(0.99)$). Cards should be mailed immediately after ballot receipt, both to the current address (which may be out of town) and the residence address 6 months before the election, in case a third party illicitly changed the address for the election.

6. If a Ballot-Level Comparison Audit is performed, document each sampled ballot sheet by snapping photos of both sides. These photos should be posted for review by the public to allow third parties to verify the results of the RLA.
7. Have livestreams accompany staff going into storage to retrieve the sampled ballots, to protect staff from false accusations.
8. Establish a program to test and certify locks, seals and security systems to minimize and detect tampering with any part of the election.²⁵

Title 33 STATE BOARD OF ELECTIONS

Subtitle 09 VOTING SYSTEMS — CERTIFICATION AND GENERAL REQUIREMENTS

Chapter 03 State Board Certification — Preliminary Submissions

Authority: Election Law Article, §§2-102(b)(4), 2-202(b), [and] 9-102(b), and 11-309 Annotated Code of Maryland

.05 Technical Data Package.

A. Required Information — In General. The technical data package shall contain all documentation previously submitted for the qualification testing of the system, including each of the following:

- (1) Hardware schematic diagrams;
- (2) Hardware theory of operation;
- (3) Customer maintenance documentation;
- (4) Operations manual;
- (5) Software source code (both in the form of a listing and in a machine-readable form on media acceptable to the evaluation agent);
- (6) Software system design;
- (7) Customer documentation;
- (8) Any known and unresolved system problems; and
- (9) When available, the report and test results of the independent testing laboratory.
- (10) Any capability for the system to ~~conduct~~ support a risk limiting audit of its tabulated results.

B.- C. (text unchanged)

Title 33 STATE BOARD OF ELECTIONS

Subtitle 11 ABSENTEE BALLOTS 11 Chapter 06 Post-Election Procedures

Authority: Election Law Article, §§2-102(b)(4), 2-202(b), [and] 9-303, and 11-309 Annotated Code of Maryland; 42 U.S.C. §§1973ff-1(h) and 1974

.01 Storage of Materials.

A. After completing canvassing procedures, a local board shall retain all voted ballots in a sealed ballot box labeled with a batch label until they are authorized to be opened and removed for an audit, recount, judicial challenge, storage, or disposal.

[A]B. Document Retention. A local board shall retain for 22 months after an election:

- (1) Absentee ballot applications;
- (2) Voted absentee ballots, both accepted and rejected, ballot images and Cast Vote Records;
- (3) Return envelopes In a different order from the ballots so that the correspondence of ballot to envelope cannot be determined; and
- (4) Certificates of assistance. [B]

C. Disposal of Unvoted Ballots.

²⁵ <https://votewell.cc/locks.html> cites different states' rules, Appel, Blaze, Halderman and many others,

- (1) A local board may dispose of any unvoted absentee ballots at least 30 days after State certification of an election or the conclusion of a recount, whichever is later.
- (2) A local board is not required to shred unvoted absentee ballots.

Title 33 STATE BOARD OF ELECTIONS

Subtitle 17 EARLY VOTING Chapter 07 Non-Voting Hours Procedures

Authority: Election Law Article, §§2-102(b)(4), 2-202(b), 10-301.1, [and] 11-301, and 11-309, Annotated Code of Maryland

.03 Processing and Storing Ballots.

- A. In accordance with COMAR 33.16, the local board may process provisional ballot applications received during the early voting period.
- B. The local board shall maintain ballots voted during early voting:
 - (1) Grouped by batch number, [early voting center;] and [(2) Grouped by day; and] [(3)] (2) Separate from ballots voted on election day, absentee ballots, and provisional ballots.

Title 33 STATE BOARD OF ELECTIONS

Subtitle 19 SAME DAY REGISTRATION AND ADDRESS CHANGES Chapter 02 Public Notice

Authority: Election Law Article, §§2-102(b)(4), 2-202(b), 3-305(d) and (e), and 3-306(d) and (e), Annotated Code of Maryland

.01 Minimum Requirements.

- A. State Board. After the close of registration for each election, but prior to election day, the State Administrator shall take appropriate measures to notify each pre-qualified voter how to find the individual's correct polling place. [When feasible, the State Administrator shall send a pre-election mailing to each pre-qualified voter, which shall include the correct polling place for the pre-qualified voter's address.]
- B. (text unchanged)